

Categories, Proofs, and Processes:

Lecture 13 Yoneda's Lemma

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5 Yoneda's Lemma

5.1 Motivation

There are lots of Hom-functors, one for every $A \in \text{Ob}(\mathcal{C})$.

... but they are all related by natural transformations.

$$t : \mathcal{C}(-, A) \rightarrow \mathcal{C}(-, B) := \{t_C : \mathcal{C}(C, A) \rightarrow \mathcal{C}(C, B)\}$$

For $f : C \rightarrow D$,

$$\begin{array}{ccc} \mathcal{C}(D, A) & \xrightarrow{\mathcal{C}(f, A)} & \mathcal{C}(C, A) \\ t_D \downarrow & & \downarrow t_C \\ \mathcal{C}(D, B) & \xrightarrow{\mathcal{C}(f, B)} & \mathcal{C}(C, B) \end{array}$$

The contravariant Hom-functor fixes the first element, the natural transformation fixes the second element.

Claim 5.1 (Ex. 55). For $g : A \rightarrow B$,

$$t := \mathcal{C}(-, g) := \{t_C = \mathcal{C}(C, g) : \mathcal{C}(C, A) \rightarrow \mathcal{C}(C, B)\}$$

$\mathcal{C}(-, g)$ is a natural transformation.

Proof.

$$\begin{array}{ccc} \mathcal{C}(D, A) & \xrightarrow{\mathcal{C}(f, A)} & \mathcal{C}(C, A) \\ \downarrow t_D & & \downarrow t_C \\ \mathcal{C}(D, B) & \xrightarrow{\mathcal{C}(f, B)} & \mathcal{C}(C, B) \end{array}$$

$$\begin{array}{ccc} h & \xrightarrow{\quad} & h \circ f \\ \downarrow & & \downarrow \\ g \circ h & \xrightarrow{\quad} & g \circ h \circ f \end{array}$$

□

Lemma 5.2 (Faux-neda). *There exists a bijection*

$$\phi : \text{Nat}(\mathcal{C}(-, A), \mathcal{C}(-, B)) \cong \mathcal{C}(A, B)$$

5.2 Yoneda's Lemma

Lemma 5.3 (Yoneda's Lemma). *For any functor $H : \mathcal{C}^{op} \rightarrow \mathbf{Set}$, and $A \in \text{Ob}(\mathcal{C})$, there exists a bijection (imagine $H = \mathcal{C}(-, B)$)*

$$\phi : \text{Nat}(\mathcal{C}(-, A), H) \cong H A$$

Proof. Take a natural transformation $t : \mathcal{C}(-, A) \rightarrow H$. We need an element $u \in H A$.

$$t_A : \mathcal{C}(A, A) \rightarrow H A$$

Let

$$u := t_A(1_A) \in H A, \quad \phi(t) = t_A(1_A)$$

For the inverse, take some $u \in H A$, and build a natural transformation t from it.

$$t_C : \mathcal{C}(C, A) \rightarrow H C$$

$$t_C(g) := H g u$$

Need to show that t is natural. $\forall f : C \rightarrow D$,

$$\begin{array}{ccc} \mathcal{C}(D, A) & \xrightarrow{\mathcal{C}(f, A)} & \mathcal{C}(C, A) \\ \downarrow t_D & & \downarrow t_C \\ & \begin{array}{ccc} g & \xrightarrow{\quad} & g \circ f \\ \downarrow & & \downarrow \\ H g u & \xrightarrow{\quad} & H f(H g u) \end{array} & \\ \downarrow & & \downarrow \\ H D & \xrightarrow{H f} & H C \end{array}$$

$$H f(H g u) = (H f \circ H g)u = H(g \circ f)u$$

since H is contravariant.

$$\phi(t) = t_A(1_A) \in H A$$

$$\psi(u) = t, \quad t_C(g) = H g u$$

$$\phi(\psi(u)) = \phi(t) = t_A(1_A) = H 1_A u = 1_{H A} u = u$$

$$\begin{aligned}
\psi(\phi(s)) &= \psi(s_A(1_A)) = t, & t_C(g) &= (H \ g)(s_A(1_A)) \\
t_C(g) &= (H \ g \circ s_A) \ 1_A \\
&= s_C \ \mathcal{C}(g, A) \ 1_A \\
&= s_C(g)
\end{aligned} \tag{1}$$

For $g : C \rightarrow A$,

$$\begin{array}{ccc}
\mathcal{C}(A, A) & \xrightarrow{\mathcal{C}(g, A)} & \mathcal{C}(C, A) \\
s_A \downarrow & & \downarrow c_A \\
H \ A & \xrightarrow{H \ g} & H \ C
\end{array}$$

□

Corollary 5.4. *Yoneda's Lemma [5.3] $\implies$ Faux-neda [5.2].*

Example 5.5.

$$\phi : \text{Nat}(\mathcal{C}(-, A), H) \cong H \ A$$

If $H = \mathcal{C}(-, B)$, then $\psi(f) : \mathcal{C}(-, A) \rightarrow \mathcal{C}(-, B)$,

$$\psi(f)_C(g) = \mathcal{C}(g, B)(f) = f \circ g$$

so $\psi(f) = \mathcal{C}(A, f)$

$$\phi(t) = \mathcal{C}(A, f)(1_A) = f$$